How green is Europe's electricity?

EU-27 generation mixes, lifecycle intensity, and the role of nuclear · 2010-2024

Scientific Visualization | OWID Energy dataset version 2026-09-10

Sample: 405 country-years (EU-27 x 2010-2024); 27/27 complete in 2024 for core electricity variables

Intensity unit: lifecycle gCO2eq / kWh generated | Electricity indicators mainly Ember via OWID

https://catalog.ourworldindata.org/garden/energy/2026-09-10/owid_energy/owid_energy.csv

Research questions

Three descriptive questions — no causal claims

Q1 Time

How did the EU-27 electricity mix change from 2010 to 2024?

Q2 Pathways

Which countries lead or lag — and via which generation portfolios?

Q3 Intensity

Do renewables alone track lifecycle intensity, or does nuclear matter?

Exploratory side question

Can portfolio features form interpretable groups? → K-means + PCA (weak silhouette expected)

Provenance · version-pinned snapshot



Observation unit

One row = one country in one year · not people, not EU aggregates

Pinned CSV

catalog.ourworldindata.org/garden/energy/2026-09-10/owid_energy/owid_energy.csv

Attribute OWID and underlying providers (Ember for electricity intensity).

Variables, units and denominators

Variables · units · denominators

Indicator	Unit / definition	Watch out
Renewable share	% of electricity generation	Not share of total final energy
Nuclear share	% of electricity generation	Separate from renewables
Low-carbon share	Renewables + nuclear (%)	The key composite for intensity
Fossil share	% of electricity generation	Coal + gas + oil electricity
Lifecycle intensity	gCO_2eq / kWh generated	Lifecycle GHGs, Ember via OWID
Generation (weight)	TWh	Weights EU averages

Cleaning and methodology

Data-processing flow (transparent, no imputation)

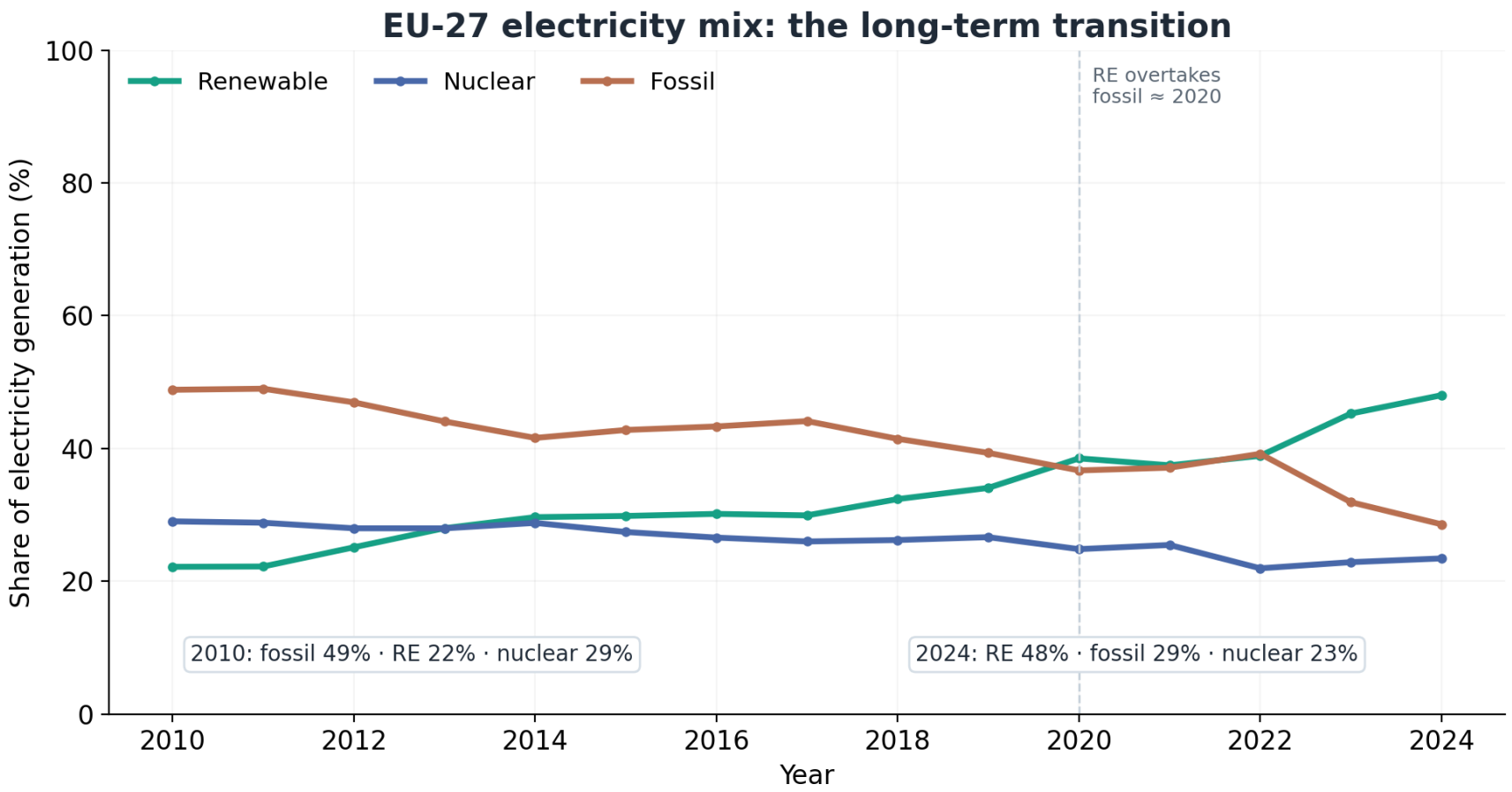


Methods used

- EU trends = generation-weighted means (country share × TWh / Σ TWh)
- Cross-section = Spearman ρ · Clusters = standardized portfolio → K-means (silhouette) + PCA 2D

How Europe's electricity mix changed

Renewables 22% -> 48% | Fossil 49% -> 29% | Nuclear 29% -> 23% (generation-weighted)



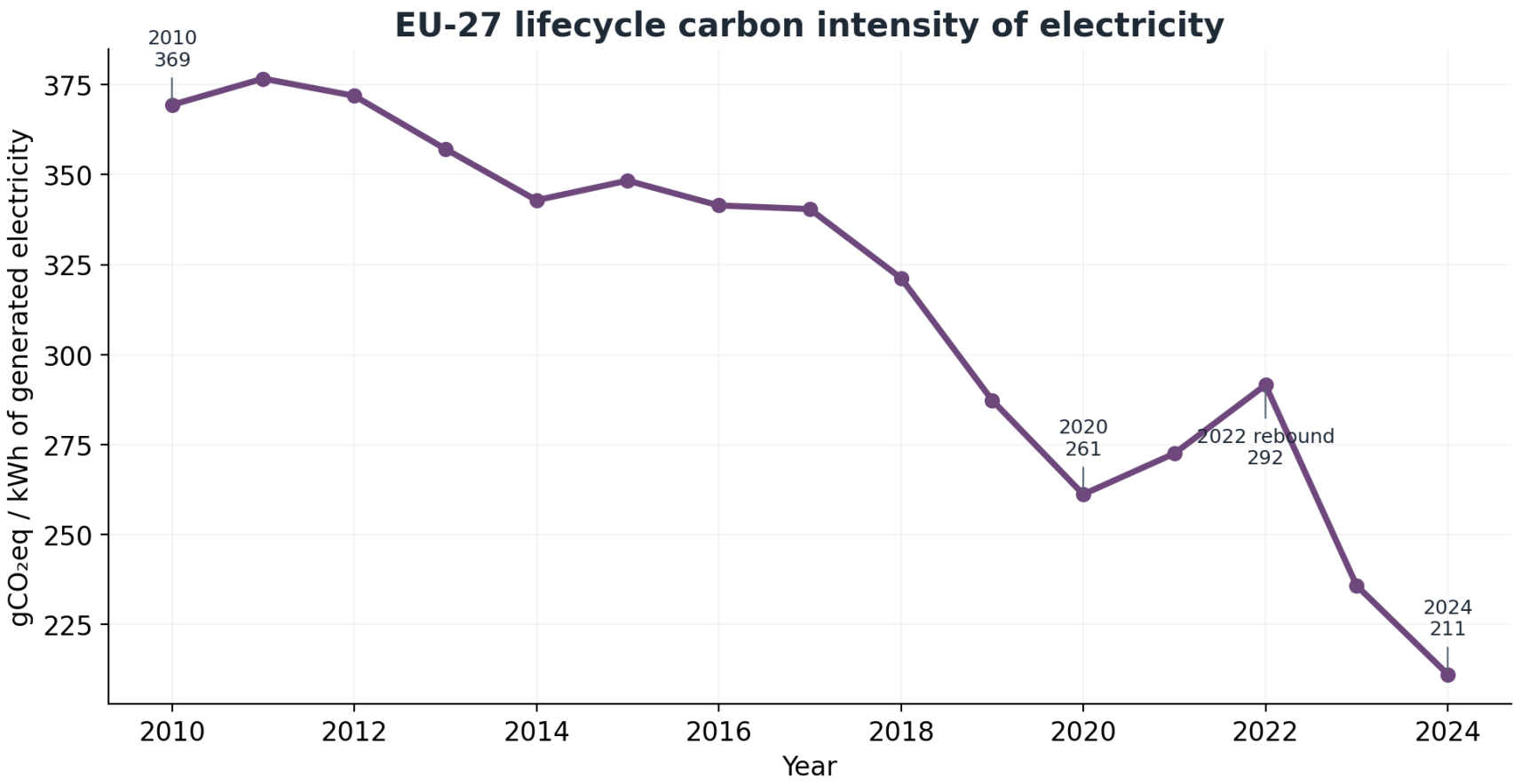
Country shares weighted by national generation (TWh). Verified from yearly_coverage_and_weighted_indicators.csv

Notice
RE overtook fossil ~2020.
Nuclear declined slowly
but remains material.

Weighted by TWh.

Carbon intensity through time

Lifecycle intensity 369 -> 211 gCO2eq/kWh (2010->2024), with 2021-22 rebound

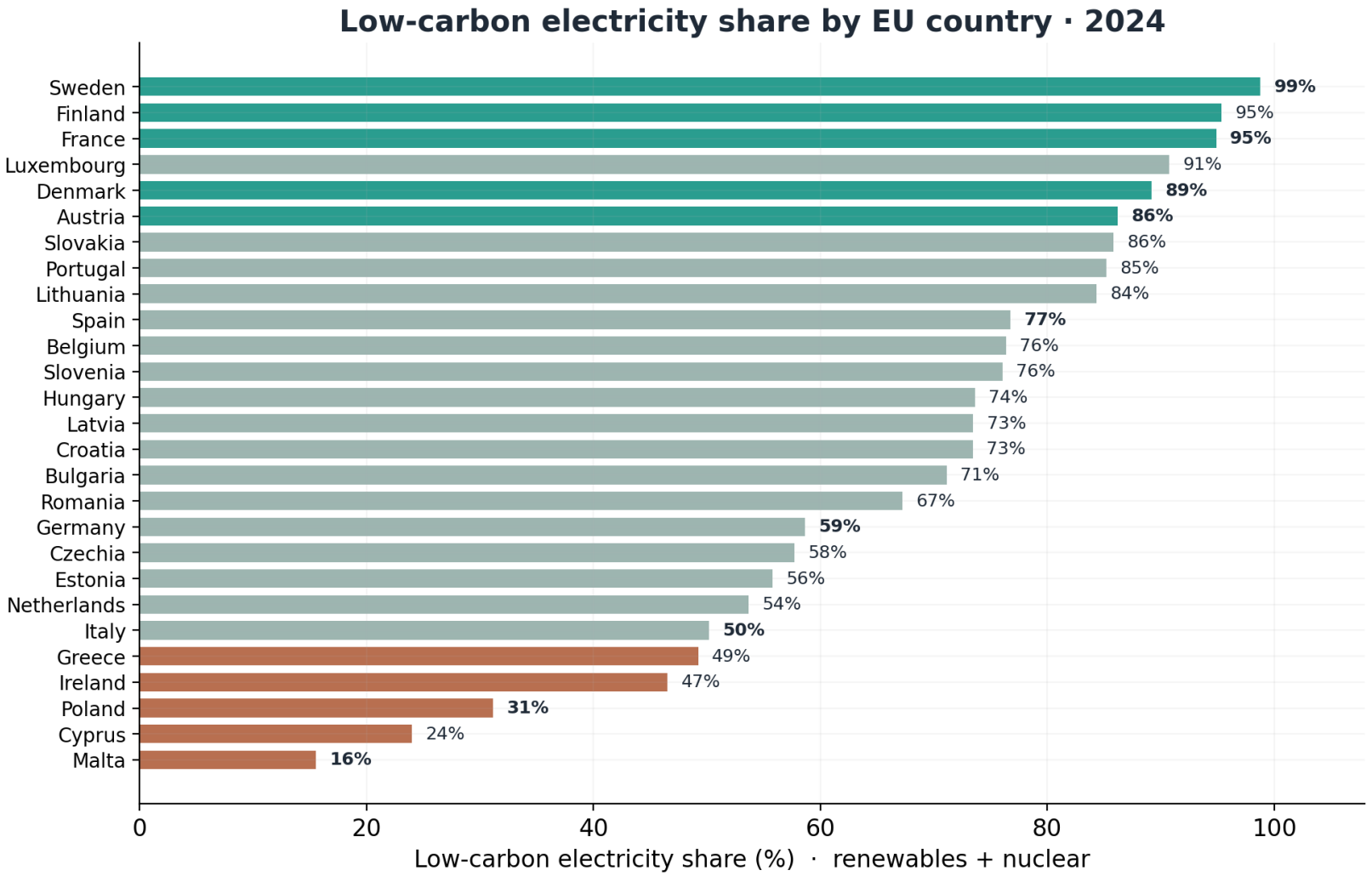


Generation-weighted · coverage 27/27 each year for this indicator set · OWID Energy 2026-09-10

Notice
~43% lower by 2024.
2022 rebound ~292.

Caveat: generation
!= consumption/imports.

Leaders and laggards · 2024

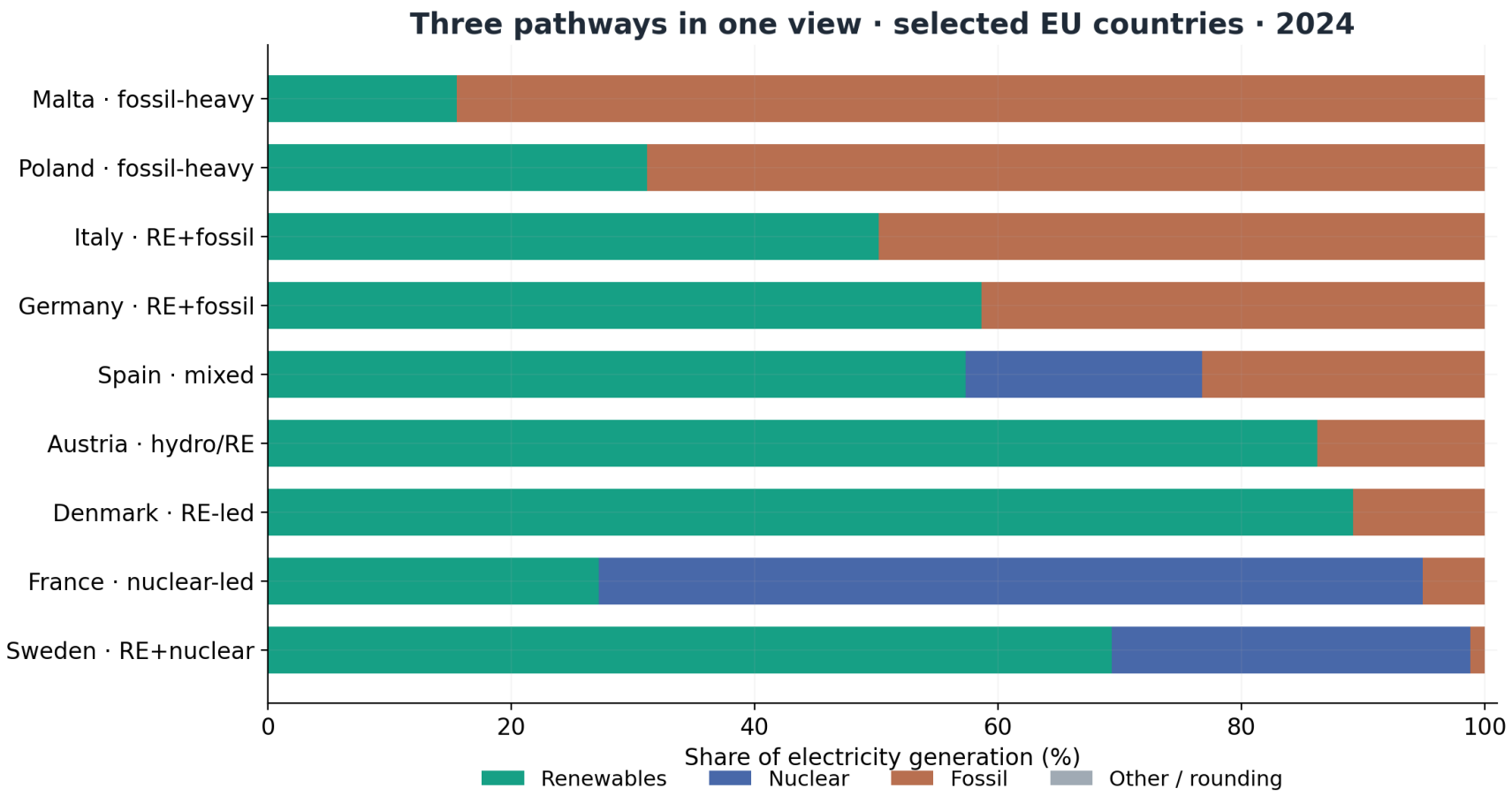


Highlighted: leaders (teal) and fossil-heavy systems (brown). Full series for all 27 countries.

Notice
Sweden ~99%, France ~95%,
Denmark ~89%.
Poland, Cyprus, Malta lag.

Full chart also in
figures/03_...png

Three pathways to electricity generation



Notice
Low fossil can mean
RE-led (DK) or
nuclear-led (FR).

Full EU-27 mix:
figures/06_...png

Presentation spotlight (9 of 27). Complete EU-27 stacked mix: figures/06_electricity_mix_by_country.png

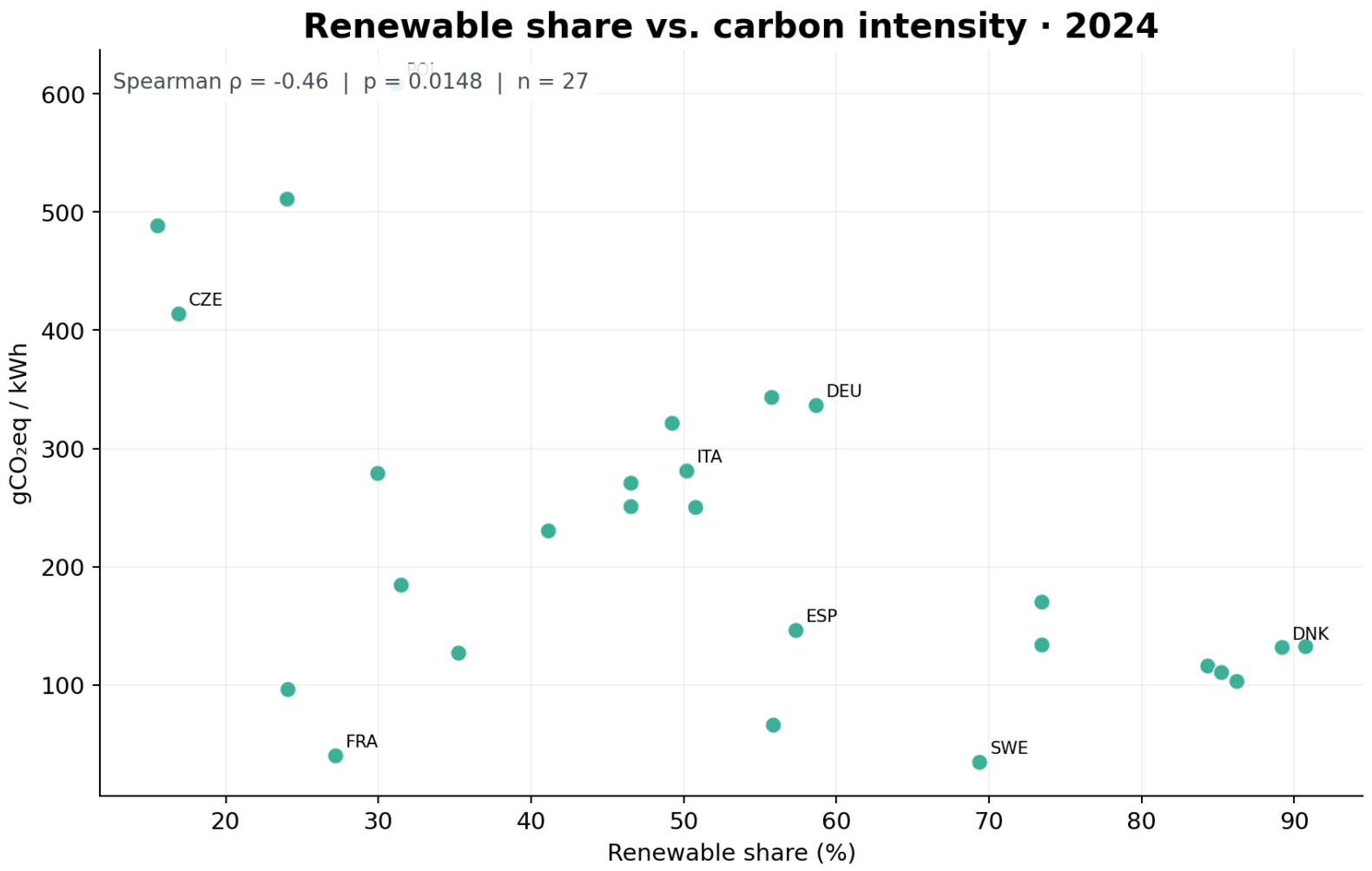
Story beat 1 · Are renewables enough?

Renewables only

Spearman rho = -0.46
p = 0.015
n = 27 (2024)

Moderate association.
France: low RE share,
very low intensity.
Poland: high intensity.

Not causal.



Cross-sectional association, not a causal effect. Countries are observations.

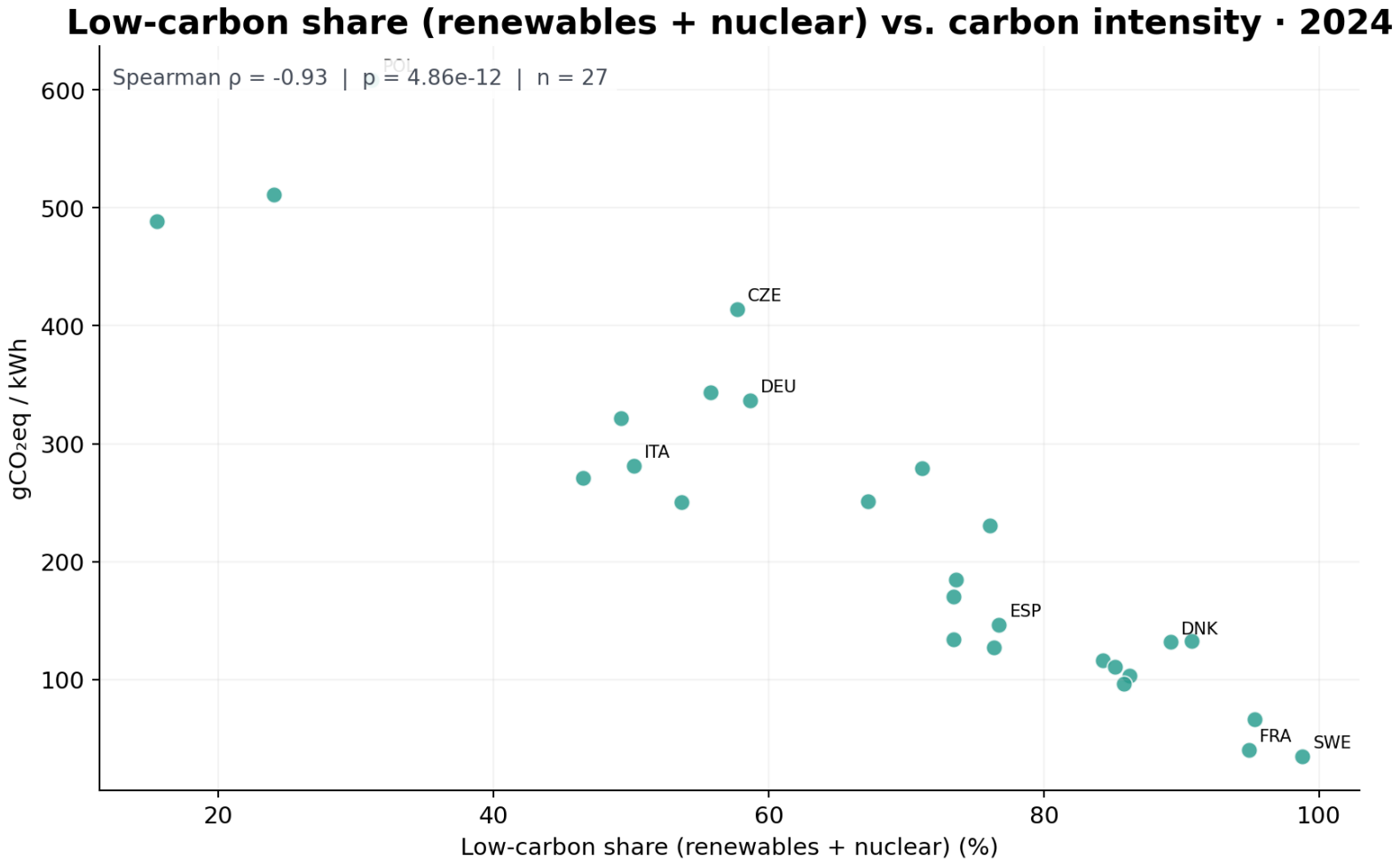
Story beat 2 · What changes when nuclear is included?

Low-carbon = RE + nuclear

Spearman rho = -0.93
p ~ 4.9e-12
n = 27 (2024)

|rho| jumps 0.46 -> 0.93.
Nuclear closes the gap
for FR / SK pathways.

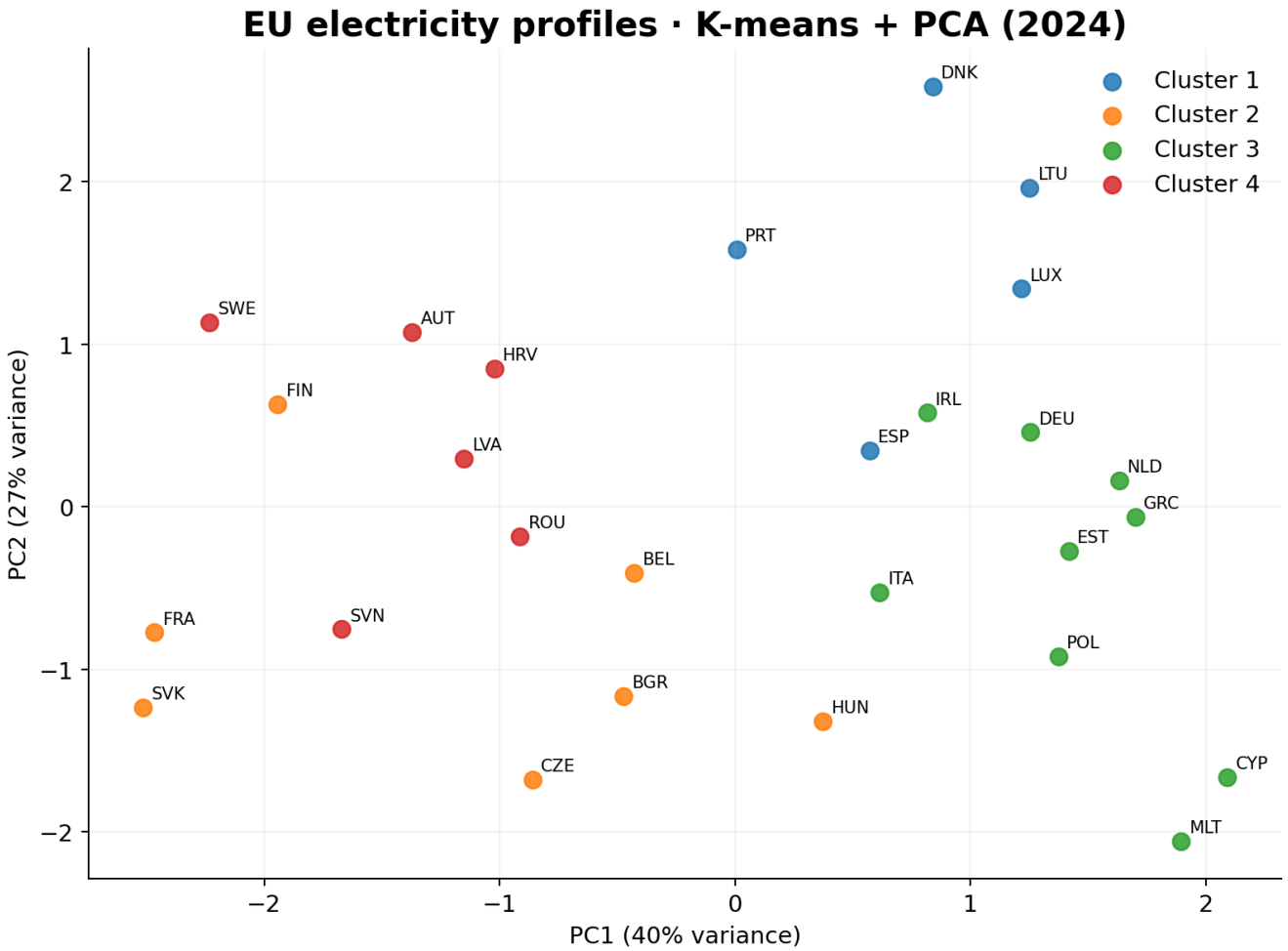
Still descriptive.



Cross-sectional association, not a causal effect. Countries are observations.

Country profiles · exploratory K-means + PCA

Best k=4 but silhouette=0.30 (WEAK) | PCA 2D ~67% variance | exploratory only



Weak separation
Silhouette ~0.30 for
all k in 2..5.

Visual aid only —
not natural regimes.

k=4, silhouette=0.30; clustering is exploratory, not a ground-truth classification.

Conclusions, limitations & sources

1. EU mix transition: RE 22->48%, fossil 49->29%, intensity 369->211 gCO2eq/kWh (2010-2024).
2. Divergent pathways: renewables-led, nuclear-led, or both among low-carbon leaders.
3. Nuclear in the story: rho -0.46 (RE) vs -0.93 (low-carbon); descriptive, not causal.

Limitations: no causality; generation != consumption/imports; lifecycle != stack-only CO2;
clustering silhouette 0.30 = weak separation.

Source: Our World in Data Energy version 2026-09-10 (electricity mainly Ember).
https://catalog.ourworldindata.org/garden/energy/2026-09-10/owid_energy/owid_energy.csv
Reproducibility: python analysis.py --year 2024 | outputs/verified_findings.md
Full EU-27 charts retained in outputs/figures/; presentation variants in figures/present/.